QPCODE

Reg.No : .....
Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) EXAMINATION OCTOBER 2025
THIRD SEMESTER
Value Addition Courses (VAC) – MG3VACPHY200
SCIENCE AND SOCIETY
(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks:50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Objective Type Questions

Answer any **25** questions

Each question carries **2** marks

1. Which of the following is arranged in the correct sequence in scientific inquiry? [U] / [CO1]
   A) Theory → Law → Hypothesis → Experiment
   B) Experiment → Hypothesis → Observation → Theory
   C) Observation → Hypothesis → Experiment → Theory
   D) Law → Experiment → Hypothesis → Theory

2. Which of the following is NOT required for a hypothesis to become a theory? [U] / [CO1]
   A) Consistent experimental support
   B) Observation and data
   C) Verification by multiple scientists
   D) Approval from religious authorities

3. What does the open-ended nature of science imply? [U] / [CO1]
   A) All current theories are wrong.
   B) Science is a never-ending process of questioning and discovery.
   C) Science ignores past knowledge completely.
   D) Scientific facts keep changing randomly.

4. Which of these can lead to the revision of a scientific theory? [U] / [CO1]
A) Religious approval
B) Political opinion
C) New evidence and discoveries
D) Public voting

5. A single counter-observation can falsify a scientific theory, no matter how many times it has been confirmed before. [U] / [CO1]
A) True
B) False
C) Cannot say
D) True only in some cases

6. Match the scientists (List 1) with the ideas (List 2) they are most associated with: [K] / [CO1]
List I (Scientists)
1. Aristotle
2. Copernicus
3. Galileo Galilei
List II (Ideas)
a. Proposed the heliocentric model
b. Supported the geocentric model (Earth-centered universe)
c. Provided telescope evidence supporting heliocentrism

A) 1–c, 2–b, 3–a
B) 1–a, 2–c, 3–b
C) 1–b, 2–a, 3–c
D) 1–b, 2–c, 3–a

7. What does the term *heliocentric* literally mean? [K] / [CO1]
A) Moon-centered
B) Sun-centered
C) Earth-centered
D) Star-centered

8. Which celestial discovery by Galileo proved that not everything revolves around Earth? [U] / [CO1]
A)Rings of Saturn
B) Moons of Jupiter
C) Sunspots
D) Craters on the Moon

9. Which of the following thinkers did NOT believe in the heliocentric theory? [K] / [CO1]
A) Copernicus
B) Ptolemy
C) Galileo Galilei
D) None of these

10. Who accurately measured the Earth's circumference using shadows and geometry? [K] / [CO1]
A)Galileo
B) Ptolemy
C) Eratosthenes
D) Kepler

11. Which unique property is typically associated with nanostructures? [U] / [CO2]
A) High surface-to-volume ratio
B) Quantum effects
C) Enhanced reactivity
D) All of the above

12. **Assertion (A):** Uranium ore as mined from the earth cannot be directly used as reactor fuel. [U] / [CO2]
Reason (R): Natural uranium contains only about 0.7% of fissile isotope U, which is insufficient for most reactors.
A) Both A and R are true, and R is the correct explanation of A.
B) Both A and R are true, but R is not the correct explanation of A.
C) A is true, R is false.
D) A is false, R is true.

13. Why does the bottom of a swimming pool look closer to the surface than it really is? [U] / [CO2]
A) Reflection of Light on the surface of water

B) Refraction of Light when it comes out of water
C) Scattering of Light in water
D) Dispersion of Light in water

14. Which mission made the first successful human landing on the Moon? [K] / [CO2]
a) Luna 9
b) Apollo 11
c) Chang'e 3
d) Chandrayaan-1

15. Which element is commonly used to make integrated circuits and microchips? [R] / [CO2]
A) Silicon
B) Gold
C) Lead
D) Zinc

16. **Assertion (A):** Recommendation systems used by Netflix or Amazon are part of AI. [U] / [CO2]
Reason (R): They study user data to suggest movies, shows, or products.

A) Both A and R are true, and R is the correct explanation of A.
B) Both A and R are true, but R is NOT the correct explanation of A.
C) A is true, but R is false.
D) A is false, but R is true.

17. **Assertion (A)**: Antibiotics are effective against all microorganisms including viruses. [U] / [CO2]
Reason (R): Antibiotics specifically target bacterial structures and processes, which viruses lack

A) Both A and R are true, and R is the correct explanation of A.
B) Both A and R are true, but R is NOT the correct explanation of A.
C) A is true, R is false.
D) A is false, R is true.

18. What is the basic unit of information in quantum computing? [U] / [CO2]

A) Byte
B) Bit
C) Qubit
D) Chip

19. What is AI bias? [U] / [CO3]
A) AI performing tasks faster than humans
B) AI showing unfair or skewed outcomes due to flawed data or algorithms
C) AI refusing to work on certain devices
D) AI replacing human jobs

20. Why are nuclear weapons considered ethically problematic? [A] / [CO3]
A) They are too expensive to maintain
B) They cause massive, indiscriminate destruction
C) They only work in certain climates
D) They cannot be transported easily

21. Which of the following is NOT a benefit of public awareness in science and technology? [U] / [CO4]
A) Better health and safety choice
B) Understanding environmental issues
C) Increased superstition and myths
D) Informed voting and policy support

22. What does ‘scientific temper’ in the Indian Constitution encourage? [U] / [CO4]
A) Blind faith and superstition
B) Rational thinking, inquiry, and evidence-based reasoning
C) Ignoring scientific developments
D) Memorizing laws of science.

23. Which of the following is a fact about eclipses? [A] / [CO5]
A) Eclipses bring bad luck or disasters.
B) Pregnant women should not step outside during an eclipse.
C) Eating food during an eclipse is harmful.
D) One should not look directly at the Sun without protection during a solar eclipse.

24. Even under normal operation, nuclear power plants emit dangerous levels of harmful radiation into the surrounding areas, endangering the people working in the vicinity. This statement is: [U] / [CO5]
A) True
B) False

25. **Assertion (A):** The Big Bang was a giant explosion. [An] / [CO5]
Reason (R): The Big Bang was actually the rapid expansion of space.
Choose the correct option:
A) Both A and R are true, and R explains A.
B) Both A and R are true, but R does NOT explain A.
C) A is false, but R is true
D) Both A and R are false.

26. Which of the following is NOT a myth about stars? [U] / [CO5]
A) Stars twinkle because they turn on and off.
B) Shooting stars are falling stars.
C) Stars are giant balls of hot gases.
D) Stars move around Earth every night.

27. **Assertion (A):** Humans evolved from monkeys we see today. [An] / [CO5]
Reason (R): Humans, apes, and monkeys share a common ancestor, and fossils show different stages of human evolution.
Choose the correct option:
A) Both A and R are true, and R is the correct explanation of A.
B) Both A and R are true, but R is NOT the correct explanation of A.
C) A is false, but R is true.
D) A is true, but R is false.

28. Which of the following is an example of a systematic error? [A / [CO6]
A) Sometimes reading a scale slightly above, sometimes below the mark
B) A weighing scale that consistently overreads by 2 g
C) Recording fewer measurements than required for accuracy
D) Small unpredictable variations in repeated readings

29. Which of the following reduces random errors in an experiment? [U] / [CO6]
A) Using a new instrument
B) Taking multiple measurements and averaging them
C) Ignoring some of the measurements
D) None of these

30. What are spurious measurements? [U / [CO6]

A) Measurements that are slightly inaccurate
B) Measurements taken with a faulty instrument
C) Readings that are wildly different from the rest
D) Measurements that have random error

[2x25 = 50]